

Training Configuration and Summary:

Dataset used: Training: rssi_odom_dataset_11apr25_isolado.parquet | Experiment: rssi_odom_dataset_12apr25_isolado.parquet

model_type: GRU
group_sec: 100
timesteps_orig: 60
batch_size: 1024
min_units: 16
max_units: 512
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
group_msec: 100
samples_per_sec: 10
forecast_horizon_sec: 4
forecast_steps: 40
timesteps: 60

Dataset Parameters:

Training: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60
Experiment: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60

Tuning:

Trials executed: 10/10

Training Performance:

Loss: 0.0034 | MAE: 0.0448

Validation Performance:

Loss: 0.0094 | MAE: 0.0774

GRU Multi-Step Training Report

Report Generated: 2025-09-20 16:16:24

Training Start: 2025-08-04_11-23-22

Training End: 2025-08-04_11-49-56

Training Duration: 26.57 minutes

Predictions Start: 2025-08-04_11-49-56

Predictions End: 2025-08-04_11-53-50

Predictions Duration: 3.90 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: relu

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 3.3154

MSE: 16.4952

RMSE: 4.0614

R2 Score: 0.5130

Multi-Step Forecast:

Forecast Steps: 40

MAE: 0.1129

MSE: 0.0181

RMSE: 0.1344

R2 Score: 0.6112

Dataset size before down-sample: 31,097 rows
Dataset size after down-sample: 3,738 rows

First 5 timestamps BEFORE:

1744464597710
1744464597720
1744464597744
1744464597757
1744464597783

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

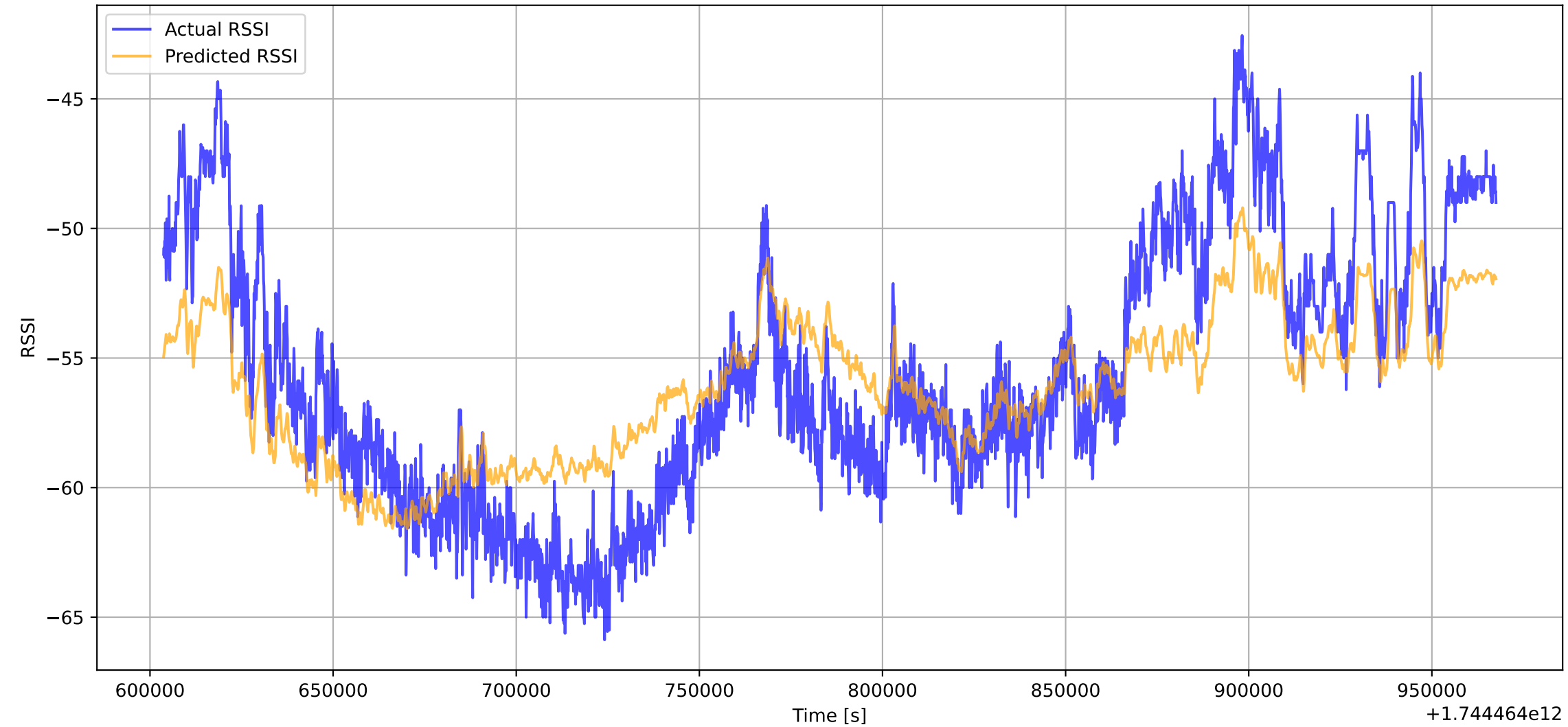
1744464597710
1744464597812
1744464597900
1744464598008
1744464598103

...

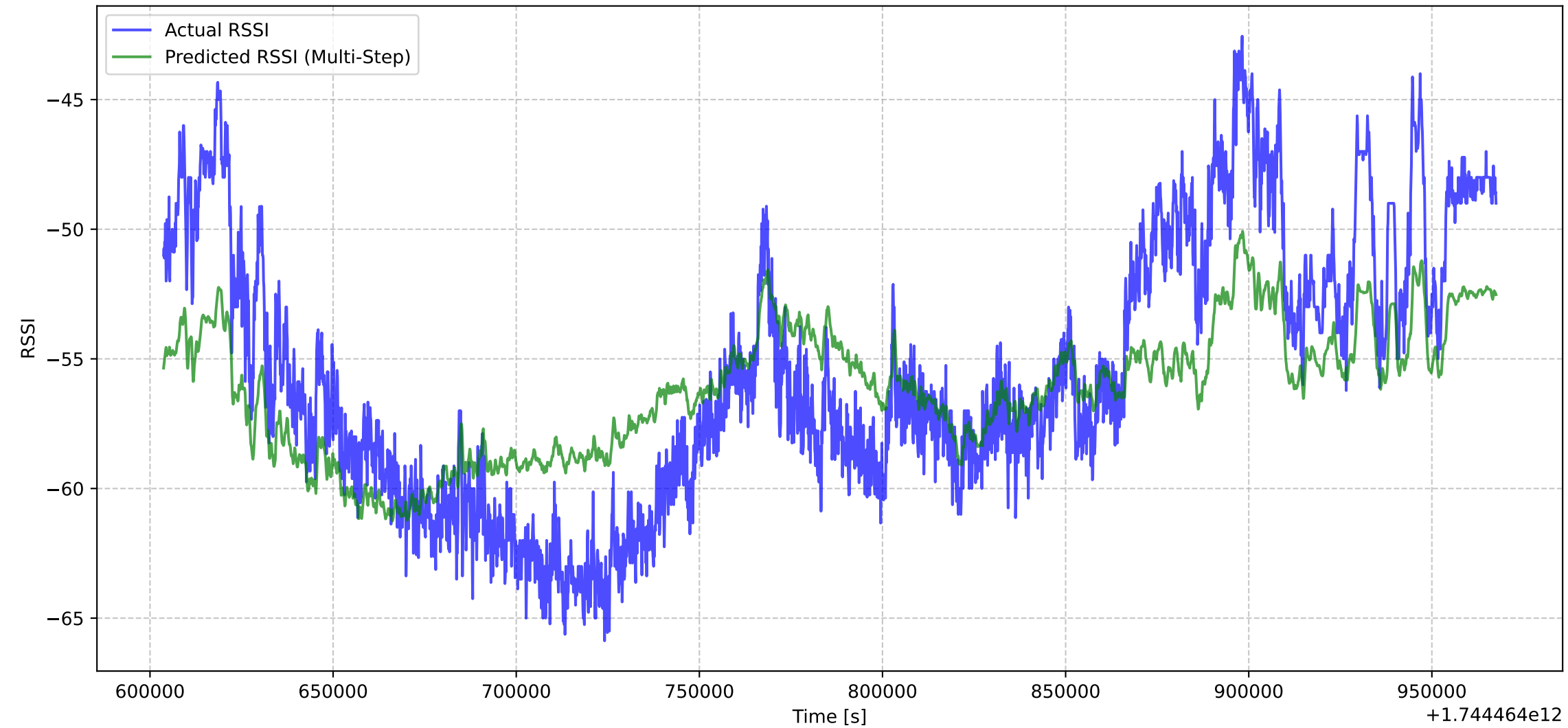
Last 5 timestamps AFTER:

1744464971004
1744464971101
1744464971209
1744464971306
1744464971402

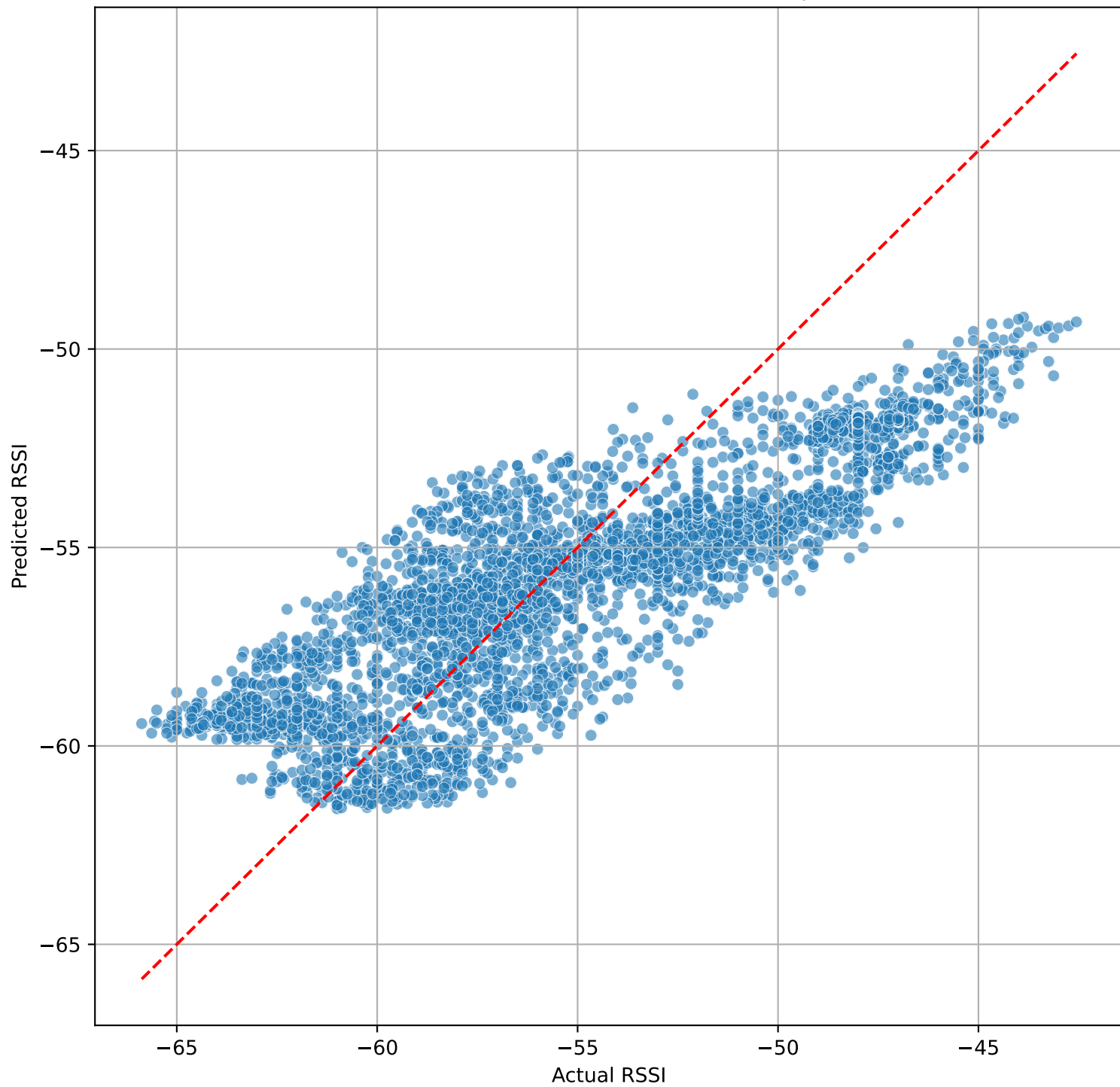
Actual vs Predicted RSSI (First Step)



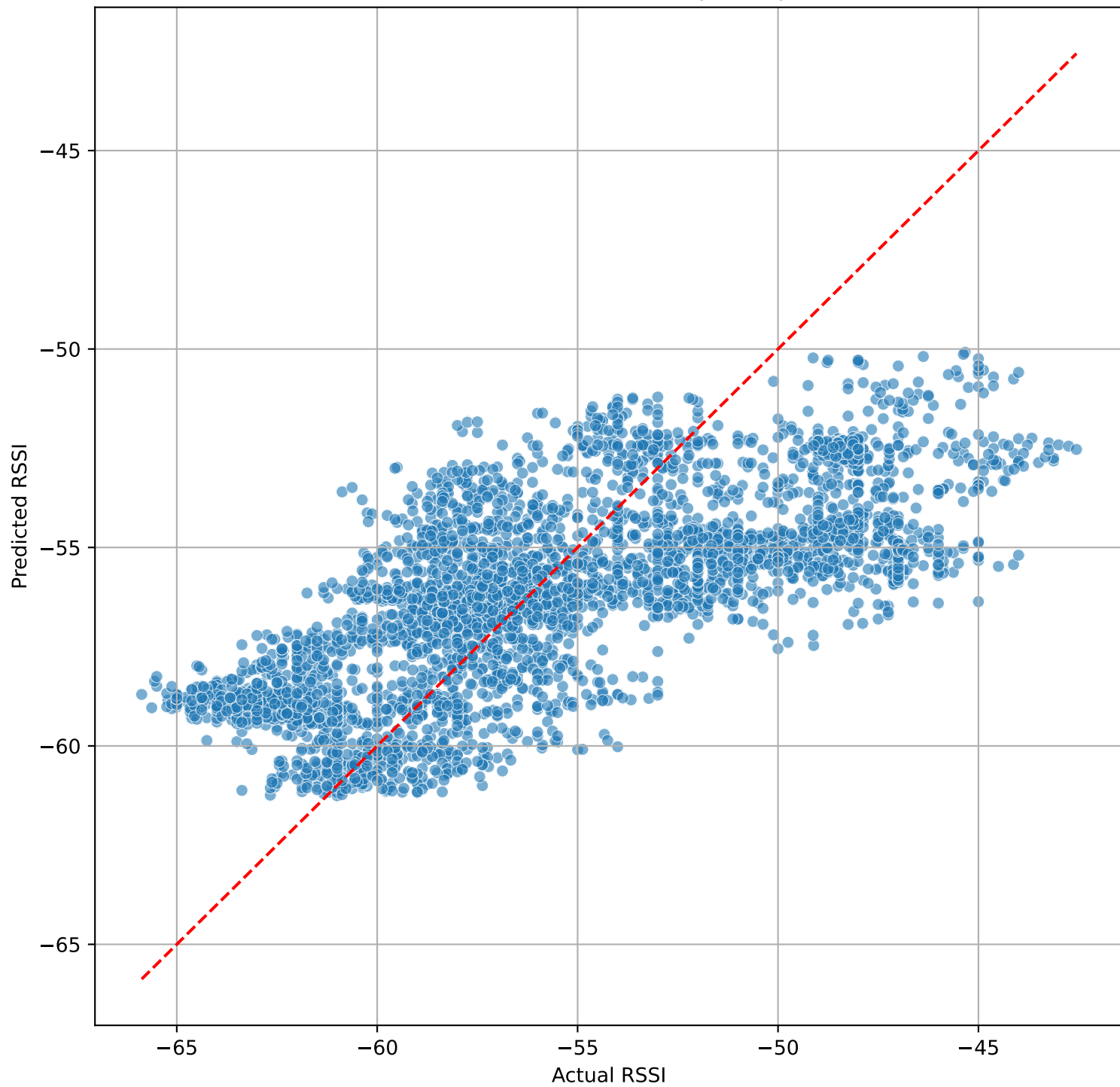
Actual vs Predicted RSSI (Multi-Step (step 40 $\approx$ 4.0 s))



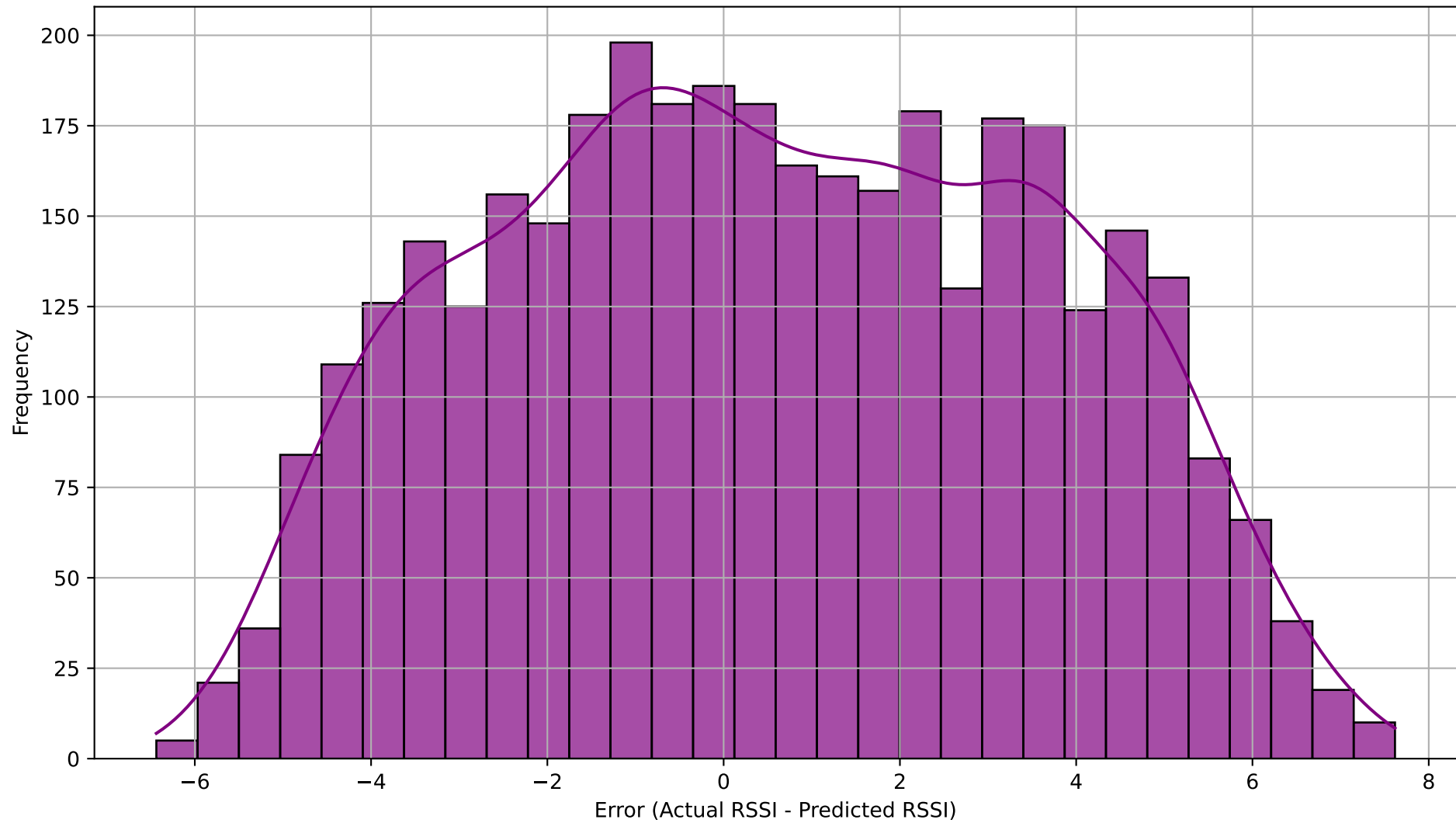
Actual vs Predicted RSSI (First Step)



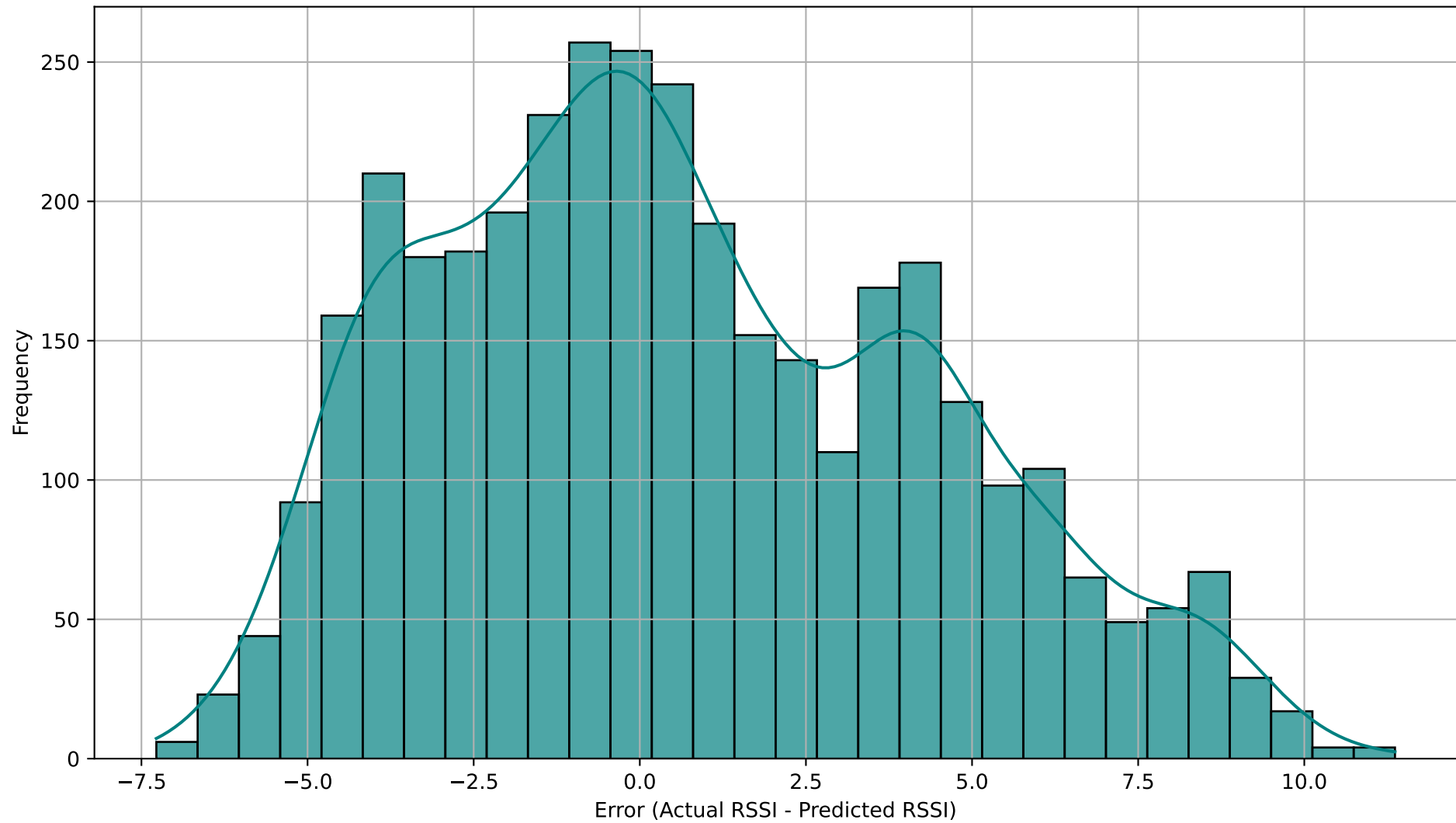
Actual vs Predicted RSSI (Last Step - step 40 $\approx$ 4.0 s)



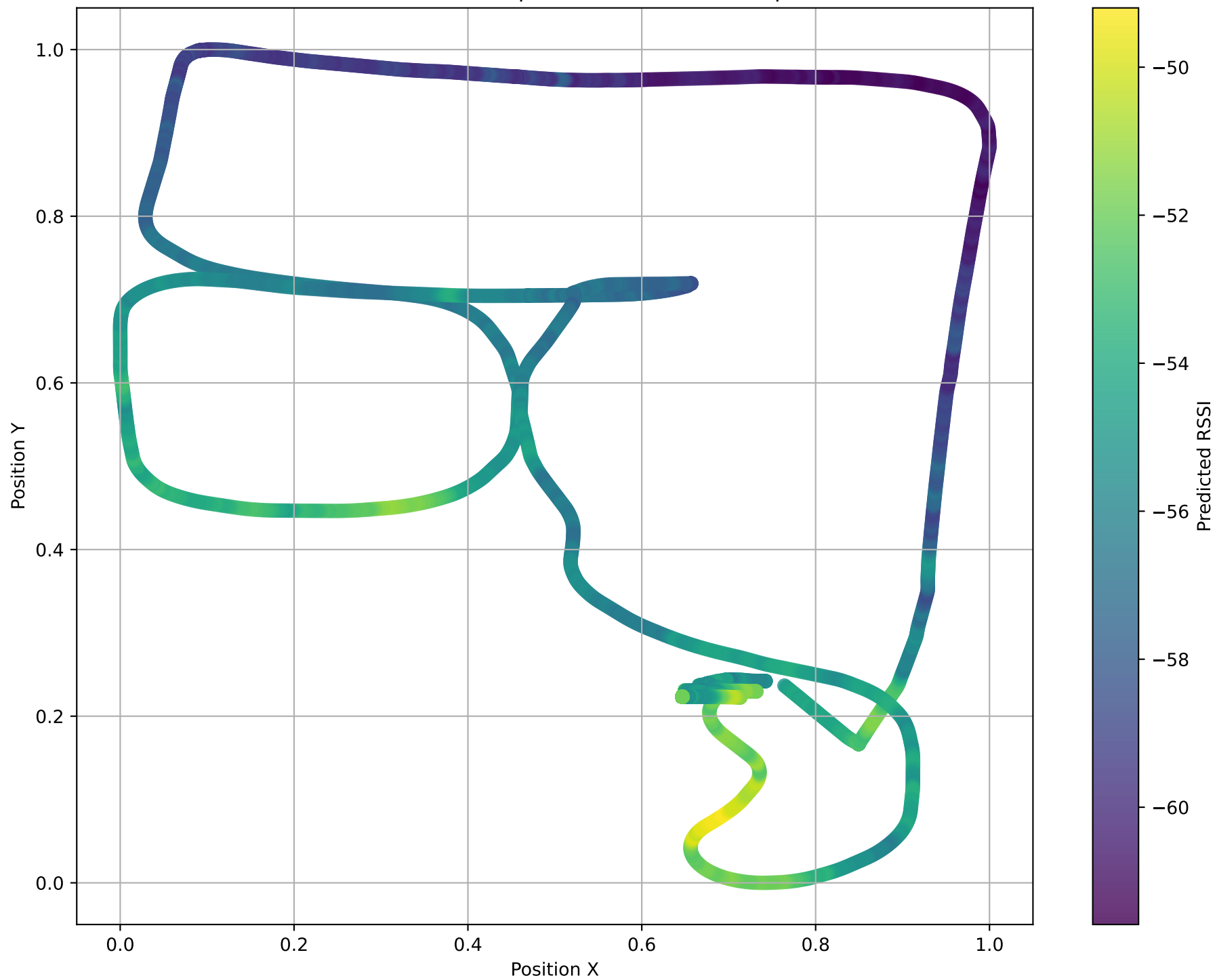
Distribution of Prediction Errors (First Step)



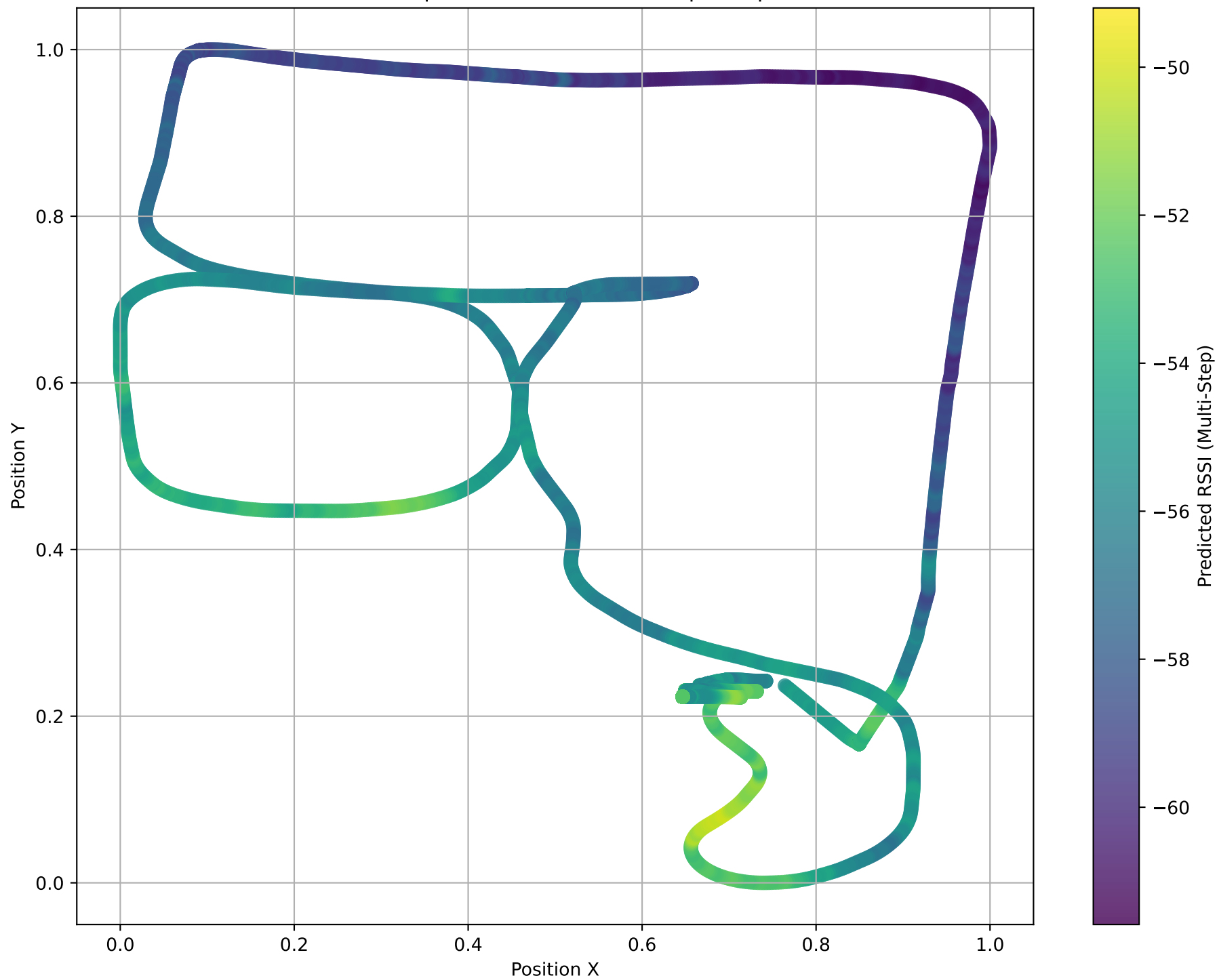
Distribution of Prediction Errors (Last Step - step 40 $\approx$ 4.0 s)



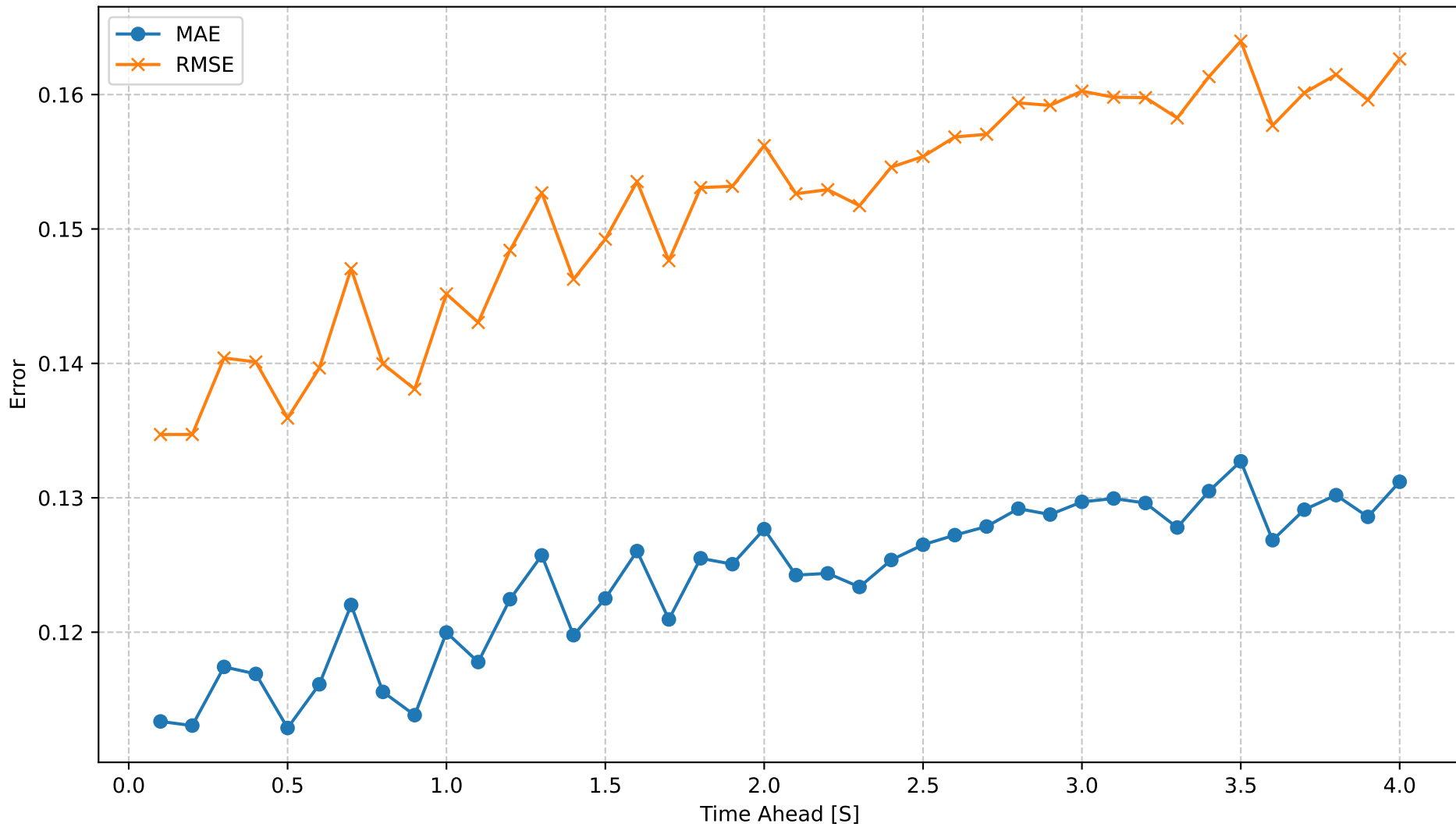
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 40 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: Training: rssi_odom_dataset_11apr25_isolado.parquet |
Experiment: rssi_odom_dataset_12apr25_isolado.parquet
Modelo: GRU
Forecast steps: 40

MAE / RMSE (dB)

- 1º passo : 0.11 / 0.13
- Mediana : 0.13 / 0.15
- 40º passo : 0.13 / 0.16

Crescimento do erro: 15.7% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 35 (0.13), 40 (0.13), 34 (0.13)

Recomendação:

O modelo é adequado para decisões de hand-over até o horizonte máximo configurado.